

Sequences and Basic Series Convergence Tests



Convergence of sequences

- 1 Determine whether the sequence $a_n = \frac{3n+1}{n}$ converges, and if so, find its limit.
 - 2 Determine whether the sequence $a_n = (-1)^n$ converges.
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Geometric series

- 3 Determine whether $\sum_{n=0}^{\infty} 3\left(\frac{1}{2}\right)^n$ converges, and if so, find its sum.
 - 4 Determine whether $\sum_{n=1}^{\infty} 2^n$ converges.
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The n th-term (divergence) test

- 5 Use the n th-term test to show $\sum_{n=1}^{\infty} \frac{n}{n+1}$ diverges.
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The integral test

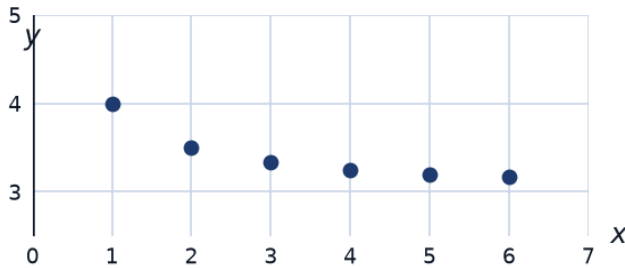
- 6 Use the Integral Test to determine whether $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges.
 - 7 Use the Integral Test to determine whether $\sum_{n=1}^{\infty} \frac{1}{n}$ converges (the harmonic series).
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The comparison and limit comparison tests

- 8 Use the Direct Comparison Test to determine whether $\sum_{n=1}^{\infty} \frac{1}{n^2+1}$ converges, comparing to $\sum \frac{1}{n^2}$.
 - 9 Use the Limit Comparison Test to determine whether $\sum_{n=1}^{\infty} \frac{2n}{n^2+3}$ converges, comparing to $\sum \frac{1}{n}$.
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Visualizing sequence convergence

- 10 Plot the first 6 terms of $a_n = \frac{3n+1}{n}$ (from the earlier question) and describe how the points behave as n increases.



Monotonic and bounded sequences

- 11 Determine whether $a_n = \frac{n}{n+1}$ is monotonic (always increasing or always decreasing), and find an upper bound.
- 12 State the Monotonic Sequence Theorem, and use it to conclude that $a_n = \frac{n}{n+1}$ converges.

The p-series test

- 13 State the p-series test for $\sum_{n=1}^{\infty} \frac{1}{n^p}$, and classify convergence for $p = 1$, $p = \frac{3}{2}$, and $p = \frac{1}{2}$.
- 14 Use the p-series test (after simplifying) to determine whether $\sum_{n=1}^{\infty} \frac{5}{n^{0.9}}$ converges.

Series review: choosing the right test

- 15 For each series, name the most efficient test to apply (do not fully evaluate):

a $\sum \frac{1}{5^n}$

b $\sum \frac{1}{n^3}$

c $\sum \frac{n+1}{2n+3}$

d $\sum \frac{1}{n^2 + 2n}$

Sequences defined recursively

- 16 A sequence is defined recursively by $a_1 = 2$, $a_{n+1} = \frac{a_n + 6}{2}$. Compute a_2 , a_3 , and a_4 , and conjecture the limit if the sequence converges.
- 17 Verify that if a recursively defined sequence $a_{n+1} = \frac{a_n + 6}{2}$ converges to a limit L , then L must satisfy $L = 6$.

Geometric series with algebraic manipulation

- 18 Find the sum of $\sum_{n=2}^{\infty} \left(\frac{2}{3}\right)^n$ (note the series starts at $n = 2$, not $n = 0$).
- 19 Express the repeating decimal $0.\overline{45}$ as a geometric series and find its exact fractional value.
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Telescoping series

- 20 Evaluate the telescoping series $\sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{1}{n+1}\right)$ by examining its partial sums.
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Multiple choice — easy 15

- 21 Find the sum of $\sum_{n=0}^{\infty} 9\left(-\frac{1}{3}\right)^n$.

- a. 12
- b. $\frac{31}{4}$
- c. $\frac{27}{4}$
- d. $\frac{27}{2}$

- 22 Find the sum of $\sum_{n=0}^{\infty} 3\left(\frac{1}{6}\right)^n$.

- a. $\frac{23}{5}$
- b. $\frac{5}{2}$
- c. $\frac{18}{7}$
- d. $\frac{18}{5}$

- 23 Find the sum of $\sum_{n=0}^{\infty} 2\left(\frac{2}{3}\right)^n$.

- a. 7
- b. $\frac{6}{5}$
- c. 6
- d. $\frac{2}{3}$

24 Find the sum of $\sum_{n=0}^{\infty} 5\left(\frac{1}{6}\right)^n$.

a. 6

b. $\frac{30}{7}$

c. 7

d. $\frac{25}{6}$

25 Find the sum of $\sum_{n=0}^{\infty} 6\left(\frac{1}{2}\right)^n$.

a. 12

b. 13

c. 4

d. 3

26 Find the sum of $\sum_{n=0}^{\infty} 9\left(\frac{2}{5}\right)^n$.

a. 16

b. 15

c. $\frac{45}{7}$

d. $\frac{27}{5}$

27 Find the sum of $\sum_{n=0}^{\infty} 3\left(\frac{1}{4}\right)^n$.

a. 4

b. $\frac{9}{4}$

c. 5

d. $\frac{12}{5}$

28 Find the sum of $\sum_{n=0}^{\infty} 7\left(\frac{1}{2}\right)^n$.

a. 15

b. $\frac{7}{2}$

c. $\frac{14}{3}$

d. 14

29 Find the sum of $\sum_{n=0}^{\infty} 3\left(-\frac{1}{2}\right)^n$.

a. 3

b. 6

c. $\frac{9}{2}$

d. 2

30 Find the sum of $\sum_{n=0}^{\infty} 1\left(\frac{1}{3}\right)^n$.

a. $\frac{3}{2}$

b. $\frac{5}{2}$

c. $\frac{3}{4}$

d. $\frac{2}{3}$

31 Find the sum of $\sum_{n=0}^{\infty} 8\left(\frac{1}{6}\right)^n$.

a. $\frac{48}{7}$

b. $\frac{48}{5}$

c. $\frac{53}{5}$

d. $\frac{20}{3}$

32 Find the sum of $\sum_{n=0}^{\infty} 1\left(\frac{2}{3}\right)^n$.

a. 4

b. 3

c. $\frac{1}{3}$

d. $\frac{3}{5}$

33 Find the sum of $\sum_{n=0}^{\infty} 4\left(\frac{2}{5}\right)^n$.

a. $\frac{20}{7}$

b. $\frac{23}{3}$

c. $\frac{12}{5}$

d. $\frac{20}{3}$

34 Find the sum of $\sum_{n=0}^{\infty} 1\left(\frac{2}{5}\right)^n$.

a. $\frac{5}{7}$

b. $\frac{3}{5}$

c. $\frac{8}{3}$

d. $\frac{5}{3}$

35 Find the sum of $\sum_{n=0}^{\infty} 7\left(\frac{1}{6}\right)^n$.

a. $\frac{35}{6}$

b. $\frac{42}{5}$

c. 6

d. $\frac{47}{5}$

Multiple choice — medium 15

36 Find $\lim_{n \rightarrow \infty} \frac{-6n+9}{7n+3}$.

a. $\frac{1}{7}$

b. $\frac{6}{7}$

c. $-\frac{7}{6}$

d. $-\frac{6}{7}$

37 Find $\lim_{n \rightarrow \infty} \frac{9n+9}{-8n+4}$.

a. $-\frac{9}{8}$

b. $\frac{9}{8}$

c. $-\frac{1}{8}$

d. $-\frac{8}{9}$

38 Find $\lim_{n \rightarrow \infty} \frac{6n-6}{-7n+2}$.

a. $-\frac{7}{6}$

b. $\frac{6}{7}$

c. $\frac{1}{7}$

d. $-\frac{6}{7}$

39 Find $\lim_{n \rightarrow \infty} \frac{-6n-5}{-2n-7}$.

a. -3

b. 4

c. 3

d. $\frac{1}{3}$

40 Find $\lim_{n \rightarrow \infty} \frac{-n-5}{6n-8}$.

a. $-\frac{1}{6}$

b. -6

c. $\frac{5}{6}$

d. $\frac{1}{6}$

41 Find $\lim_{n \rightarrow \infty} \frac{8n-7}{-8n+7}$.

a. $-\frac{1}{2}$

b. 1

c. 0

d. -1

42 Find $\lim_{n \rightarrow \infty} \frac{3n+2}{-6n-3}$.

a. -2

b. -1

c. $\frac{1}{2}$

d. $-\frac{1}{2}$

43 Find $\lim_{n \rightarrow \infty} \frac{5n-3}{9n+1}$.

a. $-\frac{5}{9}$

b. $\frac{14}{9}$

c. $\frac{9}{5}$

d. $\frac{5}{9}$

44 Find $\lim_{n \rightarrow \infty} \frac{-5n+4}{6n-2}$.

a. $-\frac{6}{5}$

b. $\frac{5}{6}$

c. $-\frac{5}{6}$

d. $\frac{1}{6}$

45 Find $\lim_{n \rightarrow \infty} \frac{-7n-9}{8n+3}$.

a. $\frac{1}{8}$

b. $\frac{7}{8}$

c. $-\frac{8}{7}$

d. $-\frac{7}{8}$

46 Find $\lim_{n \rightarrow \infty} \frac{2n+9}{8n+9}$.

a. $-\frac{1}{4}$

b. 4

c. $\frac{1}{4}$

d. $\frac{5}{4}$

47 Find $\lim_{n \rightarrow \infty} \frac{6n+7}{-7n-4}$.

a. $-\frac{7}{6}$

b. $\frac{6}{7}$

c. $\frac{1}{7}$

d. $-\frac{6}{7}$

48 Find $\lim_{n \rightarrow \infty} \frac{5n-2}{-6n-3}$.

a. $\frac{1}{6}$

b. $-\frac{6}{5}$

c. $-\frac{5}{6}$

d. $\frac{5}{6}$

49 Find $\lim_{n \rightarrow \infty} \frac{8n-6}{-6n+8}$.

a. $-\frac{4}{3}$

b. $-\frac{1}{3}$

c. $-\frac{3}{4}$

d. $\frac{4}{3}$

50 Find $\lim_{n \rightarrow \infty} \frac{-2n+5}{n+9}$.

a. -2

b. 2

c. -1

d. $-\frac{1}{2}$

Multiple choice — hard 15

51 Find the sum of $\sum_{n=2}^{\infty} 5\left(\frac{2}{5}\right)^n$ (note the series starts at $n = 2$).

a. $\frac{10}{3}$

b. $\frac{25}{3}$

c. $\frac{4}{3}$

d. $\frac{7}{3}$

52 Find the sum of $\sum_{n=1}^{\infty} 4\left(-\frac{2}{3}\right)^n$ (note the series starts at $n = 1$).

a. $\frac{12}{5}$

b. $-\frac{21}{10}$

c. $-\frac{3}{5}$

d. $-\frac{8}{5}$

53 Find the sum of $\sum_{n=1}^{\infty} 8\left(-\frac{1}{3}\right)^n$ (note the series starts at $n = 1$).

a. -2

b. 6

c. -1

d. $-\frac{5}{2}$

54 Find the sum of $\sum_{n=2}^{\infty} 2\left(\frac{2}{3}\right)^n$ (note the series starts at $n = 2$).

a. $\frac{8}{3}$

b. 4

c. 6

d. $\frac{11}{3}$

55 Find the sum of $\sum_{n=2}^{\infty} 9\left(\frac{2}{5}\right)^n$ (note the series starts at $n = 2$).

a. 15

b. $\frac{17}{5}$

c. $\frac{12}{5}$

d. 6

- 56 Use the p -series test to classify $\sum_{n=1}^{\infty} \frac{1}{n^{\frac{2}{3}}}$.
- a. Converges ($p = \frac{2}{3} > 1$)
 - b. Diverges ($p = \frac{2}{3} \leq 1$)
 - c. Diverges (terms don't approach 0, by the n th-term test)
 - d. Converges (geometric series test)

- 57 Use the p -series test to classify $\sum_{n=1}^{\infty} \frac{1}{n^{\frac{1}{2}}}$.
- a. Converges ($p = \frac{1}{2} > 1$)
 - b. Converges (geometric series test)
 - c. Diverges ($p = \frac{1}{2} \leq 1$)
 - d. Diverges (terms don't approach 0, by the n th-term test)

- 58 Use the p -series test to classify $\sum_{n=1}^{\infty} \frac{1}{n^{\frac{3}{2}}}$.
- a. Converges (geometric series test)
 - b. Converges ($p = \frac{3}{2} > 1$)
 - c. Diverges ($p = \frac{3}{2} \leq 1$)
 - d. Diverges (terms don't approach 0, by the n th-term test)

- 59 Use the p -series test to classify $\sum_{n=1}^{\infty} \frac{1}{n^2}$.
- a. Diverges ($p = 2 \leq 1$)
 - b. Converges ($p = 2 > 1$)
 - c. Converges (geometric series test)
 - d. Diverges (terms don't approach 0, by the n th-term test)

- 60 Use the p -series test to classify $\sum_{n=1}^{\infty} \frac{1}{n^{\frac{4}{3}}}$.
- Converges ($p = \frac{4}{3} > 1$)
 - Diverges (terms don't approach 0, by the n th-term test)
 - Diverges ($p = \frac{4}{3} \leq 1$)
 - Converges (geometric series test)
- 61 Use the Limit Comparison Test on $\sum_{n=1}^{\infty} \frac{4n}{6n^2 - 9}$, comparing to $\sum \frac{1}{n}$. Find $\lim_{n \rightarrow \infty} \frac{a_n}{b_n}$.
- $-\frac{2}{3}$
 - $\frac{3}{2}$
 - $\frac{2}{3}$
 - $\frac{5}{3}$
- 62 Use the Limit Comparison Test on $\sum_{n=1}^{\infty} \frac{6n}{5n^2}$, comparing to $\sum \frac{1}{n}$. Find $\lim_{n \rightarrow \infty} \frac{a_n}{b_n}$.
- $\frac{6}{5}$
 - $\frac{5}{6}$
 - $\frac{11}{5}$
 - $-\frac{6}{5}$
- 63 Use the Limit Comparison Test on $\sum_{n=1}^{\infty} \frac{5n}{7n^2 + 4}$, comparing to $\sum \frac{1}{n}$. Find $\lim_{n \rightarrow \infty} \frac{a_n}{b_n}$.
- $-\frac{5}{7}$
 - $\frac{7}{5}$
 - $\frac{12}{7}$
 - $\frac{5}{7}$

64 Use the Limit Comparison Test on $\sum_{n=1}^{\infty} \frac{7n}{3n^2+9}$, comparing to $\sum \frac{1}{n}$. Find $\lim_{n \rightarrow \infty} \frac{a_n}{b_n}$.

a. $\frac{7}{3}$

b. $-\frac{7}{3}$

c. $\frac{3}{7}$

d. $\frac{10}{3}$

65 Use the Limit Comparison Test on $\sum_{n=1}^{\infty} \frac{n}{5n^2+6}$, comparing to $\sum \frac{1}{n}$. Find $\lim_{n \rightarrow \infty} \frac{a_n}{b_n}$.

a. $\frac{6}{5}$

b. $\frac{1}{5}$

c. 5

d. $-\frac{1}{5}$